

FieldPulse AI

Complete Local Machine Setup, Execution & User Operations Manual

Document Version: 2.4 · **Generated:** September 13, 2026 · **Architecture:** Next.js 14 App Router + FastAPI Python 3.12 + MongoDB Atlas

1. Executive Overview & System Architecture

FieldPulse AI is an enterprise construction telemetry and schedule intelligence platform designed for large capital projects, civil infrastructure, and oil & gas pipeline packages. It bridges the gap between raw physical site progress (photos, voice notes, QR spools, GPS tags) and executive milestone schedules using multi-signal AI matching (CLIP vision, faster-whisper, and Mistral NLP).

Service	Component	Host Port	Access URL / Notes
Frontend App	Next.js 14 (App Router / PWA)	3000	http://localhost:3000
Backend API	FastAPI (Python 3.12+)	8000	http://localhost:8000/docs (Swagger)
Database	MongoDB 7.0 (Local or Atlas)	27017	Configured via MONGO_URI

2. Prerequisites & Environment Requirements

The platform supports both Windows (PowerShell/CMD), macOS (Terminal), and Linux (Bash). You can choose between the **1-Command Docker setup** (fastest) or **Manual setup**:

Path	Required Software	Recommended Version	Verification Command
Option A: Docker (Recommended)	Docker Desktop (with Compose)	Docker 24.0+ / Compose v2+	docker compose version
Option B: Manual	Node.js & npm Python MongoDB (or Atlas)	Node 18.17+ or 20+ Python 3.11 or 3.12+ Mongo 6.0+	node -v python --version mongod --version

3. Method A: 1-Command Docker Setup (Recommended)

Docker Compose is pre-configured in the repository to spin up the persistent MongoDB container, the FastAPI Backend with automatic seed migration, and the Next.js Frontend with a single command.

Step 1: Clone the GitHub repository

```
git clone https://github.com/bhushan806/FieldPulse-AI.git
cd FieldPulse-AI
```

Step 2: Build & launch all containers

```
docker compose up --build
```

■ What Docker automates for you:

- Automatically creates internal bridge network fieldpulse_network.
- Provisions persistent volume mongo_data so your projects and evidence survive restarts.
- Executes python seed_db.py on startup to auto-provision all 5 demo persona accounts.
- Configures Next.js frontend proxy to communicate with backend on port 8000.

To stop containers: Press Ctrl + C or run docker compose down in another terminal.

To run in background: Use docker compose up -d.

4. Method B: Manual Local Setup (Without Docker)

If running natively on your operating system without Docker, follow these two terminal steps:

Terminal 1 — Backend Service Setup (FastAPI Python):

```
# 1. Navigate to backend
cd backend

# 2. Create and activate virtual environment
# Windows (PowerShell):
python -m venv venv
.\venv\Scripts\activate
# macOS / Linux:
python3 -m venv venv
source venv/bin/activate

# 3. Install dependencies
pip install -r requirements.txt

# 4. Setup environment file
# Windows: copy .env.example .env | Mac/Linux: cp .env.example .env

# 5. Seed initial projects and demo accounts
python seed_db.py

# 6. Start the FastAPI server
uvicorn app.main:app --reload --port 8000
```

Terminal 2 — Frontend Service Setup (Next.js 14):

```
# 1. Navigate to frontend
cd frontend

# 2. Install Node dependencies
npm install

# 3. Setup local environment file
# Windows: copy .env.local.example .env.local | Mac/Linux: cp .env.local.example .env.local

# 4. Launch Next.js dev server
npm run dev
```

5. Complete Role Authentication & Credentials Directory

All demo shortcut auto-fills have been removed to preserve enterprise privacy. Users must manually enter credentials into the standard login portals:

Role Persona	User Name	Email / Phone	Password / OTP	Portal Path	Target Dashboard
Project Manager	Priya Sharma	pm@fieldpulse.dev	Password123!	/login-office	/pm/dashboard
HQ Admin	Ankit Verma	hq@fieldpulse.dev	Password123!	/login-office	/hq/portfolio
Compliance Auditor	Sonal Mehta	auditor@fieldpulse.dev	Password123!	/login-office	/hq/portfolio
Site Engineer	Ramesh Kumar	+91 98765 43210	Mock OTP: 123456 (Shown on UI)	/login-engineer	/engineer/home

Platform Admin	Super Admin	admin@fieldpulse.dev	AdminPass123!	/login-admin	/admin/dashboard
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6. Operational Workflow & Feature Testing Guide

Follow this 4-step walkthrough to test the full lifecycle from field evidence capture to schedule recalculation:

Step 1: Site Engineer Field Capture (Mobile / PWA)

1. Navigate to `http://localhost:3000/login-engineer`.
2. Enter phone: +91 98765 43210 and click **Send OTP**.
3. Notice the **Mock OTP banner** appears displaying 123456. Type 123456 into the OTP box and click **Verify & Login**.
4. On the Engineer Home screen, click **Capture Evidence** (`/engineer/capture`).
5. Upload a construction photo (or select a sample from `demo_photos/`), record an audio note, or scan a pipe QR code.
6. Click **Submit Capture**. The evidence is cryptographically SHA-256 hashed and synchronized to the backend.

Step 2: Multimodal AI Matching Engine

1. When submitted, the backend AI fusion engine evaluates the evidence against active WBS schedule activities.
2. **CLIP Vision Model** assesses visual alignment (e.g., steel pipe trenching vs. excavation).
3. **Whisper Audio** transcribes spoken notes into structured construction observations.
4. **Spatial Proximity** validates that device GPS coordinates match the project geo-fence.
5. Items with high confidence ($\geq 80\%$) auto-verify; borderline items stage in the Project Manager review queue.

Step 3: Project Manager Review & Approval

1. Open `http://localhost:3000/login-office` and sign in with: `pm@fieldpulse.dev` / `Password123!`.
2. Open the **Review Queue** (`/pm/review-queue`) to inspect pending engineer captures.
3. Click any submission card to view side-by-side photo proof, audio transcript, and AI feature breakdown.
4. Click **Approve**. The activity progress percentage instantly updates.

Step 4: AI Site Time Machine & Live S-Curve Sync

1. Navigate to **Schedule** (`/pm/schedule`) or **Time Machine** (`/pm/time-machine/[id]`).
2. View the dynamic **S-Curve**: the actual progress curve recomputes in real time against the planned baseline.
3. Inspect the **Downstream Impact Graph** to observe how verified milestones mitigate critical path delay.
4. Auditor persona (`auditor@fieldpulse.dev`) can verify the SHA-256 cryptographic chain of every approval.

7. Frequently Asked Questions & Troubleshooting

Issue / Symptom	Root Cause	Resolution Step
MongoDB Connection Timeout	Atlas IP Whitelist blocking current network connection.	In MongoDB Atlas Console, go to Network Access and add IP <code>0.0.0.0/0</code> (Allow anywhere).
Port 8000 or 3000 in use	Another local process is occupying the default port.	Stop conflicting processes or start uvicorn with <code>--port 8001</code> and adjust <code>NEXT_PUBLIC_API_URL</code> .
Dark Mode Text Invisibility	Hardcoded white background on form elements.	Resolved in Core v2.4. Inputs use <code>bg-transparent</code> and adapt automatically to theme.
Mock OTP Not Received via SMS	<code>OTP_PROVIDER=mock</code> is active by default.	No physical SMS is sent. The 6-digit OTP displays directly in the UI helper banner (default 123456).

FieldPulse AI Engineering Documentation
Repository: <https://github.com/bhushan806/FieldPulse-AI> · Maintained for Production & Competition Release.